Package netlist generation rule
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Ramschip, Inc.

Quality management Div.

5F 506.43, Changeop-ro, Sujeong-gu, Seongnam-si, Gyeonggi-do, South Korea (13449)

Tel. +82.31.622.7790

RS-COP-02-I-02 Revision history

Rev. no	Rev. date	Simple description of changed contents	Author	Post
1.0	2025.03.13	Newly established.	Ronnie.Lee	QM

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1. Purpose

The purpose of this document is to define the rules for generating package netlists used in semiconductor package design.

Establishing standardized netlist generation guidelines ensures consistency, enhances design quality, and improves efficiency in automation, verification, and data exchange processes.

The package netlist serves as a critical reference for IC bonding, PCB interfacing, and simulation model development.

2. Apply scope

This rule applies to the following areas:

- All new or updated **IC package designs**
- **Package substrate** layout and connectivity definition
- Netlist generation for **signal integrity (SI)** and **power integrity (PI)** simulations
- Reference for **assembly**, **test development**, and manufacturing documentation

Note: For advanced packaging technologies such as wafer-level packaging (WLP) or system-in-package (SiP), separate or supplementary rules may apply.

3. Definition of terms

3.1 Netlist

A list describing electrical connections between pins or pads in a circuit.

3.2 Package netlist

A netlist defining connections between IC die pads and external package pins/balls.

3.3 Pin name

A unique identifier assigned to a pin or ball on the die or package.

3.4 Net name

A logical signal name representing a group of electrically connected nodes.

3.5 Ball map

A matrix layout showing the positions and labels of external package balls.

3.6 Bump pad

A bonding location on the IC die that connects to the package.

3.7 Pin name

A unique identifier assigned to a pin or ball on the die or package.

3.8 Verification

A logical signal name representing a group of electrically connected nodes.

4. Product code generation rule

4.1 IP Code generation rule

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(Category)_(Product symbol)(Product code)_(Process code) – All lower case

※ In the case of product symbol for sub-license IP, the defined name from Samsung Foundry should be used.

(Category)	(Product symbol)	(Product code)-4digit	(Process code)
(ramschip s-license IP) rs_ (ramschip own-IP) rc_	dcphy dphy cphy hsstptx hdptx etphy (ethernet phy) mtpphy aphy - -	28nm: 1000 ~ 1999 14nm : 2000 ~ 2999 8nm : 3000 ~ 3999 5nm : 5000 ~ 5999 4nm : 6000 ~ 6999	ln28lpp ln14lpu ln08lpu ln05lpe ln04lpp - - -

4.2 IP Code update rule

(Category)_(Product symbol)(Product code)(rev symbol)_(Process code) – All lower case

(Category)	(Product symbol)	(Product code)-4digit	(rev symbol)	(Process code)
(ramschip s-license IP) rs_ (ramschip own-IP) rc_	dcphy dphy cphy hsstptx hdptx etphy (ethernet phy) mtpphy aphy - - -	28nm: 1000 ~ 1999 14nm : 2000 ~ 2999 8nm : 3000 ~ 3999 5nm : 5000 ~ 5999 4nm : 6000 ~ 6999	a,b,c,d,e,f ------- ▶Except 'r'	ln28lpp ln14lpu ln08lpu ln05lpe ln04lpp - - -

4.3 Product code generation & revision rule

(Category)_(Product code #1)(Product code #2)(Version) – All upper case

Ex) RS5011A / RS5012A

(Category)	(Product code #1)	(Product code #2)	(Version)
(ramschip product) rs_	50: MIPI APHY Family ICs	11: RX 12: TX 13: RX/TX	A,B,C,D -----Z

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4.4 Design Kit naming rule

(IP name)_(Classification#1)_ (Classification#2)_(YYMMDD)

※ Classification #2: Generate in uppercase / Configuration #1: Optionable

ex) rs_hsstpech_ln14lpp_v1p0_master_FE_230323

(IP name)	Classification #1	Classification #2	YYMMDD
Defined IP name	master slave	FE BE DOC	230323

5. Relevant document

Standard	Document code
Development process	RS-COP-01

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